Q10: The Jagged Frontier

Why 'simple' is a human notion the model never learned

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The puzzle

Student question (#18)
Why can models solve complex problems yet still make simple mistakes?

It explains relativity, then miscounts letters in “strawberry”.

Human vs model competence

- Humans: competence is *correlated* (hard $\Rightarrow$ easy follows).
- Models: a **jagged frontier** — sharp, unpredictable variation.

"Hard" and "easy" are human notions the model never learned.

Root cause 1: the objective

- Trained to predict next token, not to be uniformly competent.
- Capability tracks **training frequency**, not logical difficulty.
- Relativity prose is abundant; exact long arithmetic is rare.

Root cause 2: tokenization

- Model sees sub-word tokens, not letters or digits.
- Counting r's needs letter structure it doesn't represent.
- Number chunks break positional arithmetic.

Predictable representation artifacts, not "stupidity".

Root cause 3: no built-in checking

- One forward pass, no scratchpad, no verification.
- Humans re-check; a base model doesn't unless prompted / given a tool.
- Same model often fixes it the instant you say "think step by step".

Predict the jaggedness

Strong where: well-represented, error-tolerant, pattern-like.
Weak where: exact symbols, long chains, sub-token info, rare knowledge.

Pair brittle regimes with reasoning, tools, and verifiers.